Engineering Mathematics - I · Unit_02 · Unit 2

DRAFT — UE26MA141A · Unit_02 · Unit 2 · Particular integral of standard functions - Type 4

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Particular integral of standard functions - Type 4

1 Particular integral of standard functions - Type 4

Study focus: Particular integral of standard functions - Type 4. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000126 · EV0000127

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we are learning a method or a rule to find the particular integral of function of the type a par ax into v where v is any function of x, that is v.

Key details

- We are learning a method to find the particular integral of product of two functions in which one function is exponential function e bar ax.
- Then sine a x plus b or cos e x plus b and algebraic function of the type x to the par n.
- 1 over f of d of e par e x into b is e par x into 1 over f of d plus a of b.
- See e par ax can be taken outside the operator by replacing b by d plus a in f of d.
- What is A here?
- A is the coefficient of x in the exponential function.
- This is the rule used to find the particular integral of product of two functions in which one function is x.
- But in this session, we are solving problems only on type 4, wherein the function is e par ax into v.
- Find the general solution of d2y by dx2 minus 2 dy by dx plus 4y is equal to e to the pi.
- The auxiliary equation is m square minus 2m plus 4 is equal to 0.
- Find out the roots of this quadratic equation.
- By using the quadratic formula, we get the roots as 1 plus R minus.
- Minus root 3i alpha plus r minus i beta alpha is 1 beta is root 3.
- See look at this.
- This is e par x into b where v is equal to cos x.
- E power x can be taken outside the operator by replacement by d plus a, a is one.
- d plus 1 in this e f of d.

- Just replace all d'present in f of d by d plus 1.
- This simplifies to e power x into cos x by d square plus 2d plus 1 minus 2d minus 2 plus 4.
- into cos x by d square plus 3.
- Which rule are we using?
- The second one, cos a x by f of d.
- Cos x by minus 1 square that is minus 1 plus 3, which is e power x into cos x by 2.
- This should be by p.
- Therefore, the general solution is y is equal to yc plus y p.
- This is y p.
- Particular integral when you have product of two functions in which one function is exponential function.
- Product of 3.

Example or application from the lecture

- The given function is of the type e par a x into b.
- Let us consider a problem.
- x into cos x because the given equation is a non-homogeneous LD with constant coefficient, I need to find the complementary function and particular integral.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000126 · EV0000127 · EV0000128 · +1 source units in pack.json

Core explanation · 05:00–09:59

It is given by x into e power x cos x by d square minus 2d plus 1.

Key details

- into x cos x by d plus 1 the whole square minus 2 into d plus 1 plus 1 which simplifies to this is a plus b the whole square d square plus 2 d plus 1 minus 2 d minus 2 plus 1.
- Plus 2 gets cancelled with minus 2 I get a very simple function of D in the denominator x cos x by d square.
- I know that 1 over d square of any function of x which I call it as capital X is.
- Two times integration of capital X with respect to X.
- I need to integrate this X into cos X with respect to X two times without writing the constant of integration, that's very important.
- without writing the constant of integration.
- I use integration by parts what is integration by parts integral of u into b is u integral b minus integral b into differential of u which you had studied in your 12th standard so it is e to the power x into 1 over d of let me first simplify this is it is x into integration of sine x is minus cos x minus integral of.
- Integration of sin x is minus cos x into differentiation of x with respect to x is 1 dx.
- It simplifies to e to the power x into 1 over d of minus x cos x minus into minus is plus and integration of cos.
- cos x is just sine x.
- We are not writing the constant of integration here.

- How do I integrate this minus x cos x?
- Again by using the same integration by parts minus x into integration of cos x is sine x minus integration of cos x is sine x.
- That sine x when you integrate you get minus cos x.
- Integration of this just sine x is minus cos x.
- This is the particular integral and the general solution is Yc plus YPC.
- Please look here to integrate this x is sine x.
- product of two functions.
- I have used integration by parts.
- You have used integration by parts two times.
- When to integrate x into sin x here and again to integrate this minus of x into cos x.
- This is must y is equal to yc plus y p.
- If the question is to find the general solution of the differential equation.
- Find the general solution of d2 y by dx to minus three dy by dx plus two into y is equal to x into e power 3x plus sine 2x here right hand side I have sum of two functions one is sine a x plus b form for the thirst first term I need to use this rule now for.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000130 · EV0000131 · EV0000132 · +1 source units in pack.json

Core explanation · 10:00–14:59

Now let us first write the equation in our notation form, which is d square minus 3d plus 2 of y is equal to x into e power 3x plus sin 2x.

Key details

- Now to find the complementary function, I need to first write the auxiliary equation.
- This is m square minus 3m plus 2.
- equal to 0, find the roots.
- The roots are 1 comma 2.
- So the complementary function is c 1 e to the power 1x plus c2 e to the power 2x y because the roots are real and distinct.
- It is x e to the power 3x plus sin 2x divided by d square minus 3d plus 2.
- First split this into two terms.
- X e to the power 3x by f of d plus sine 2x by f of d where f of d is d square minus 3d plus 2.
- I'll call the first term as particular integral 1 and the second term as particular integral 2, where particular integral 1 is the first term and particular integral 2 is the second term sine 2x by t square minus 3t plus 2.
- Let me first simplify.
- particular integral one and particular integral two and then substitute back in this now to find particular integral one.
- What is it?
- Product of two functions in which one function is e par three x.
- E par three x can be taken outside the operator by replacing b by b plus three in this f of b, so which is [ASR low confidence; check audio].

- x divided by b plus 3 the whole square minus 3 into d plus 3 plus 2.
- X divided by D square plus 3d plus 2.
- Now the next question is how do I simplify this?
- Has many steps.
- What is the first step?
- I need to take the lowest d degree term to get the expression in the form 1 plus 5d or 1 minus 5 d.
- of D where phi F D is D square plus three D by two.
- Take this to the numerator and expand using binomial expansion.
- 1 plus d square plus 3d by 2 whole raise to minus 1 of x.
- Though the tur though the expansion has infinite.
- number of terms because the function on which it has to be operated is x to the power 1.
- The first order derivative of x to the power 1, that is d of x to the power 1 itself is a constant 1.
- When you differentiate it 2, 3, n times the derivatives would be 0.
- So I want only the first derivative d to the power 1.
- I have neglected.
- All higher degree terms.
- In fact, it is 1 minus b square plus 3d by 2 to the power 1.
- Plus d square plus 3d by 2 whole raise to 2.
- I have not considered those terms.
- In the second term also, I have not taken d square by 2.
- I have considered only 3d by 2.
- I need to find particular integral 2 also.
- It's sine 2x divided by d square minus 3d plus 2.
- Which rule I am using over here?
- Rule 2, type 2 type.
- Replace d square by minus into 2 square that is minus 4.
- Retain D term as it is.
- And 2x by 3d plus 2.
- I have a D term.
- I can apply the rule only when I have D square term.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000134 · EV0000135 · EV0000136 · +1 source units in pack.json

Core explanation · 15:00–19:59

Convert this d term into d square term by multiply and dividing by its conjugate.

Key details

- What is the conjugate?
- 3d minus 2.
- 2x, which simplifies to 3d minus 2 sin 2x divided by a minus b into a plus b, a square minus b square, 9 d square minus 4.

- Two different ways.
- See I can first apply the operator 1 over 9 d square minus 4 unsine 2x and.
- Apply the operator 3d minus 2 on the resultant function of x.
- First itself you apply the operator 3d minus 2 on sine to x and then you apply the operator 1 over 9d square minus 4 on the resultant function.
- Taken this first 1 over 9 d square minus 4 of sine to x.
- How do I simplify?
- Replace d square by minus into a square.
- What is that minus into a square?
- Minus into 4.
- It is 1 by 4 t into 3d minus 2 of sine 2x.
- Multiply 3d of sine 2x minus 2 of sine 2x.
- What is d of sine 2x?
- It is 2 cos 2x.
- It is PI1 plus PI2, and because the question is to find the general solution, y is equal to complementary function plus particular integral.

Example or application from the lecture

- This is how we find the particular integral when the given function is x into b, where b is a function of x.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000138 · EV0000139 · EV0000140 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Particular integral of standard functions - Type 4 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/006_Particular_integral_of_standard_functions_-_Type_4__603379d034480436.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Particular integral of standard functions - Type 4 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/006_Particular_integral_of_standard_functions_-_Type_4__603379d034480436.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 4 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/006_Particular_integral_of_standard_functions_-_Type_4__603379d034480436.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9988; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 4 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/006_Particular_integral_of_standard_functions_-_Type_4__603379d034480436.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
2. 2 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
3. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
4. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
5. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
6. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00030	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...
SRC00031	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00032	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...
SRC00033	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...

C Revision and retrieval practice